Abhishek Ahirrao

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Education

San José State University, M.S. Computer Science, GPA 3.71/4.0 · Student Assistant, CFETI — Expected May 2027
University of Mumbai, B.E. Computer Engineering, GPA 8.4/10 — Jul 2021

Skills

Cloud: AWS (Lambda, ECS Fargate, EC2, Step Functions, S3, VPC, IAM, RDS, Redshift, DMS, DynamoDB, Bedrock), GCP
Infrastructure & Delivery: Terraform, CloudFormation, Docker, Kubernetes, Jenkins, GitHub Actions, Vault, Trivy
Observability & Reliability: CloudWatch, Prometheus, Grafana, Loki, SLOs, incident response, disaster recovery
Languages & Data: Python, SQL, Bash, C++/CUDA; PostgreSQL, Redshift, Snowflake, DynamoDB
Certifications: AWS Solutions Architect Associate, AWS Developer Associate, HashiCorp Terraform Associate
Awards: Winner, AWS x Datadog GenAI Hackathon (2025); 1st place of ∼50 teams, Composio x Redis Hackathon (2025)

Experience

Quantiphi · *Mumbai, India* — Jan 2022 – Jun 2025
SDE-Platform II — Jan 2024 – Jun 2025

- Built a self-service infrastructure platform adopted by 50 data engineers across 5 data platforms: a Python CLI and a Claude-on-Bedrock interface that turned plain-English requests into pre-approved Terraform and CloudFormation stacks, with approval gates and TTL teardown. Cut platform-team requests ∼85% and dev spend ∼18%.
- Architected the migration of an Informatica-on-EC2 ETL platform to Lambda and Dockerized ECS Fargate services orchestrated by Step Functions, ingesting change data from 8+ sources across 5 countries into Redshift. Authored the design and cost model (projected 70% lower compute), mentored 2 juniors, and handed it to a 5-person support team.
- Split day-long full loads into restartable 2-hour batches so failures reran only the missing batch, holding a T+1 freshness SLA on 95%+ of runs as volume doubled from 300 GB to 600 GB/day.
- Enforced policy-as-code guardrails on 120 CloudFormation and Terraform stacks across 3 accounts with Control Tower, SCPs and Lambda checks; non-compliant stacks rolled back automatically.
- Served on-call for 6 production services and consolidated 3 environments into one Grafana dashboard with CloudWatch symptom-based alerts, cutting MTTR from 3 hours to 10 minutes.
- Shipped a major release every 2 weeks and 1–2 tuning releases a week via Jenkins with two-layer rollback: CloudFormation reverted failed stack updates, and failed post-deploy checks redeployed the last known-good template and image.

SDE-Platform I — Jan 2022 – Dec 2023

- Authored CloudFormation and Terraform for 40 interdependent stacks per environment, and built the platform network: segmented VPC subnets, NACLs, security groups and IAM isolation between team workloads.
- Built and operated 30 daily ETL Lambdas per environment with run-locking, health checks and control-table backups, deployed through Jenkins pipelines with Trivy scanning.
- Ran 5 types of disaster-recovery drill every 4 months; restored a failed production RDS instance in about an hour.

Projects

Phantom: IaC Drift Reconciliation SaaS | *Python, AWS Lambda, Bedrock, DynamoDB, Terraform* · *GitHub*

- Hackathon-winning prototype detected Terraform and CloudFormation drift; led a 3-person rebuild into a multi-account SaaS that reads customer CloudTrail via STS and fans out one Lambda per drifted stack.
- Generated fixes with Bedrock, validated them with cfn-lint and opened GitHub PRs; load tests on 10 live stacks scaled 1:1 with zero errors and isolated Bedrock P99 latency as the bottleneck.

Cache-Aware Scale-In for LLM Serving (M.S. project, in progress) | *Python, Kubernetes, vLLM, llm-d*

- Built a Kubernetes controller that scores each vLLM replica's KV-cache value and sets pod-deletion-cost so scale-in removes the coldest replica; benchmarking TTFT and cache hit rate on Llama 3 8B (3 A100 replicas).

Duplicate-Safe Order Ledger Service | *Python, FastAPI, PostgreSQL, ECS Fargate, RDS, k6* · *GitHub*

- Deployed an idempotent order API on ECS Fargate behind an ALB with RDS Postgres and SSM-stored secrets; same-key retries replay the stored response in one transaction, and k6 load tests sustained 250 req/s at 13 ms p95 with zero errors.